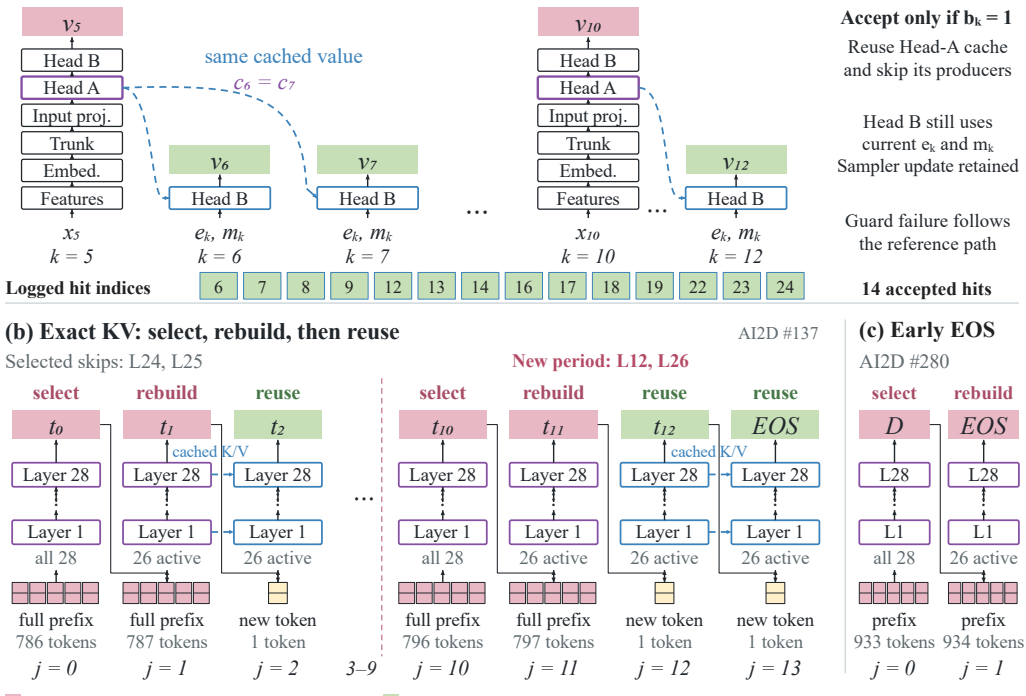

(a) CADD: cached state bypasses its producers
Recorded dispatch events
v5
Head B
Head A
Input proj.
Trunk
Embed.
Features
same cached value
c6 = c7
v6
Head B
v7
Head B
v10
Head B
Head A
Input proj.
Trunk
Embed.
Features
v12
Head B
x5
k = 5
ek, mk
k = 6
ek, mk
k = 7
x10
k = 10
ek, mk
k = 12
Logged hit indices
6 7 8 9 12 13 14 16 17 18 19 22 23 24
Accept only if bk = 1
Reuse Head-A cache and skip its producers
Head B still uses current ek and mk
Sampler update retained
Guard failure follows the reference path
14 accepted hits
(b) Exact KV: select, rebuild, then reuse
AI2D #137
Selected skips: L24, L25
New period: L12, L26
select
rebuild
reuse
t0
t1
t2
cached K/V
Layer 28
Layer 1
all 28
26 active
full prefix
786 tokens
j = 0
full prefix
787 tokens
j = 1
new token
1 token
j = 2
3–9
t10
t11
t12
EOS
full prefix
796 tokens
j = 10
full prefix
797 tokens
j = 11
new token
1 token
j = 12
new token
1 token
j = 13
full-prefix / fresh computation
cache reuse
cached state
(c) Early EOS
AI2D #280
D
EOS
L28
L1
prefix
933 tokens
j = 0
prefix
934 tokens
j = 1
2 steps, no KV reuse